DA3409+DA3410 — Deep Learning (+ lab)

Tier	A
Phase (order)	4
Do this first	DA2400, DA1000, DA1001
Estimated effort	12-14 weeks
Unlocks	DA3451, DA3452, DA6401, DA6800

Roadmap

1. MLP and backprop; optimisers.
2. CNNs; RNNs; transformers.
3. PyTorch lab: reproduce each model.

Video courses (all links checked on YouTube on 2026-10-22)

🎓 Main course: follow it start to end

- Deep Learning (100+ videos). NPTEL *Deep Learning* (Mitesh Khapra, IIT Madras).

🎥 Other full lecture series (pick one if the main one doesn't click, or for topics it skips)

- Deep Learning (68 videos). IIT Madras BS *Deep Learning*: newer recording of the IITM course.
- Neural Networks: Zero to Hero (10 videos). Karpathy *Neural Networks: Zero to Hero*: build backprop, MLPs and GPT in code. Best for the lab (DA3410).
- MIT 6.S191: Introduction to Deep Learning (90 videos). MIT 6.S191 *Introduction to Deep Learning* (official, updated yearly): fast survey.

⚡ Short videos for intuition and revision

- Neural networks (10 videos). 3Blue1Brown *Neural networks*: watch first.

Study plan (week by week)

Lecture numbers are positions in the main playlist above. Click a number to open that video.

Week	Lectures	What to do
1	1–18	History, MP neuron, perceptron, convergence proof. Code a perceptron.
2	19–34	Sigmoid neurons, gradient descent, feedforward nets, backpropagation. Write backprop for an MLP in NumPy and check with numerical gradients.

Week	Lectures	What to do
3	35–43	Optimisers: momentum, NAG, SGD/mini-batch, learning-rate schedules, Adagrad/RMSProp/Adam. Compare them on MNIST.
4	44–51, 33	Linear algebra recap, PCA, SVD.
5	52–57	Autoencoders (denoising, sparse, contractive). Train one in PyTorch.
6	58–69	Bias-variance, regularisation, augmentation, early stopping, dropout.
7	70–74	Pre-training, activations, initialisation, batch norm.
8	75–86	Word representations: word2vec, GloVe.
9	87–100	CNNs and visualising them. Train a CNN on CIFAR-10.
10-12	rest of playlist (after 100)	RNNs, LSTMs, seq2seq, attention, transformers: open the full playlist. Then Karpathy's GPT video (alt).

▼ **All lectures in *Deep Learning* (100+ videos)**

1. Deep Learning - Course Introduction
2. Deep Learning(CS7015): Lec 1.1 Biological Neuron
3. Deep Learning(CS7015): Lec 1.2 From Spring to Winter of AI
4. Deep Learning(CS7015): Lec 1.3 The Deep Revival
5. Deep Learning(CS7015): Lec 1.4 From Cats to Convolutional Neural Networks
6. Deep Learning(CS7015): Lec 1.5 Faster, higher, stronger
7. Deep Learning(CS7015): Lec 1.6 The Curious Case of Sequences
8. Deep Learning(CS7015): Lec 1.7 Beating humans at their own games (literally)
9. Deep Learning(CS7015): Lec 1.8 The Madness (2013-)
10. Deep Learning(CS7015): Lec 1.9 (Need for) Sanity
11. Deep Learning(CS7015): Lec 2.1 Motivation from Biological Neurons
12. Deep Learning(CS7015): Lec 2.2 McCulloch Pitts Neuron, Thresholding Logic
13. Deep Learning(CS7015): Lec 2.3 Perceptrons
14. Deep Learning(CS7015): Lec 2.4 Error and Error Surfaces
15. Deep Learning(CS7015): Lec 2.5 Perceptron Learning Algorithm
16. Deep Learning(CS7015): Lec 2.6 Proof of Convergence of Perceptron Learning Algorithm
17. Deep Learning(CS7015): Lec 2.7 Linearly Separable Boolean Functions
18. Deep Learning(CS7015): Lec 2.8 Representation Power of a Network of Perceptrons
19. Deep Learning(CS7015): Lec 3.1 Sigmoid Neuron
20. Deep Learning(CS7015): Lec 3.2 A typical Supervised Machine Learning Setup
21. Deep Learning(CS7015): Lec 3.3 Learning Parameters: (Infeasible) guess work
22. Deep Learning(CS7015): Lec 3.4 Learning Parameters: Gradient Descent
23. Deep Learning(CS7015): Lec 3.5 Representation Power of Multilayer Network of Sigmoid Neurons
24. Deep Learning(CS7015): Lec 4.1 Feedforward Neural Networks (a.k.a multilayered network of neurons)
25. Deep Learning(CS7015): Lec 4.2 Learning Paramters of Feedforward Neural Networks (Intuition)

26. Deep Learning(CS7015): Lec 4.3 Output functions and Loss functions
27. Deep Learning(CS7015): Lec 4.4 Backpropagation (Intuition)
28. Deep Learning(CS7015): Lec 4.5 Backpropagation: Computing Gradients w.r.t. the Output Units
29. Deep Learning(CS7015): Lec 4.6 Backpropagation: Computing Gradients w.r.t. Hidden Units
30. Deep Learning(CS7015): Lec 4.7 Backpropagation: Computing Gradients w.r.t. Parameters
31. Deep Learning(CS7015): Lec 4.8 Backpropagation: Pseudo code
32. Deep Learning(CS7015): Lec 4.9 Derivative of the activation function
33. Deep Learning(CS7015): Lec 6.6 PCA : Interpretation 3
34. Deep Learning(CS7015): Lec 4.10 Information content, Entropy & cross entropy
35. Deep Learning(CS7015): Lec 5.1 & Lec 5.2 Recap: Learning Parameters: Guess Work, Gradient Descent
36. Deep Learning(CS7015): Lec 5.3 Contours Maps
37. Deep Learning(CS7015): Lec 5.4 Momentum based Gradient Descent
38. Deep Learning(CS7015): Lec 5.5 Nesterov Accelerated Gradient Descent
39. Deep Learning(CS7015): Lec 5.6 Stochastic And Mini-Batch Gradient Descent
40. Deep Learning(CS7015): Lec 5.7 Tips for Adjusting Learning Rate and Momentum
41. Deep Learning(CS7015): Lec 5.8 Line Search
42. Deep Learning(CS7015): Lec 5.9 Gradient Descent with Adaptive Learning Rate
43. Deep Learning(CS7015): Lec 5.9 (Part-2) Bias Correction in Adam
44. Deep Learning(CS7015): Lec 6.1 Eigenvalues and Eigenvectors
45. Deep Learning(CS7015): Lec 6.2 Linear Algebra : Basic Definitions
46. Deep Learning(CS7015): Lec 6.3 Eigenvalue Decompositon
47. Deep Learning(CS7015): Lec 6.4 Principal Component Analysis and its Interpretations
48. Deep Learning(CS7015): Lec 6.5 PCA : Interpretation 2
49. Deep Learning(CS7015): Lec 6.6 (Part-2) PCA : Interpretation 3 (Contd.)
50. Deep Learning(CS7015): Lec 6.7 PCA : Practical Example
51. Deep Learning(CS7015): Lec 6.8 Singular Value Decomposition
52. Deep Learning(CS7015): Lec 7.1 Introduction to Autoncoders
53. Deep Learning(CS7015): Lec 7.2 Link between PCA and Autoencoders
54. Deep Learning(CS7015): Lec 7.3 Regularization in autoencoders (Motivation)
55. Deep Learning(CS7015): Lec 7.4 Denoising Autoencoders
56. Deep Learning(CS7015): Lec 7.5 Sparse Autoencoders
57. Deep Learning(CS7015): Lec 7.6 Contractive Autoencoders
58. Deep Learning(CS7015): Lec 8.1 Bias and Variance
59. Deep Learning(CS7015): Lec 8.2 Train error vs Test error
60. Deep Learning(CS7015): Lec 8.2 (Part-2) Train error vs Test error (Recap)
61. Deep Learning(CS7015): Lec 8.3 True error and Model complexity
62. Deep Learning(CS7015): Lec 8.4 L2 regularization
63. Deep Learning(CS7015): Lec 8.5 Dataset augmentation
64. Deep Learning(CS7015): Lec 8.6 Parameter sharing and tying
65. Deep Learning(CS7015): Lec 8.7 Adding Noise to the inputs

66. Deep Learning(CS7015): Lec 8.8 Adding Noise to the outputs
67. Deep Learning(CS7015): Lec 8.9 Early stopping
68. Deep Learning(CS7015): Lec 8.10 Ensemble Methods
69. Deep Learning(CS7015): Lec 8.11 Dropout
70. Deep Learning(CS7015): Lec 9.1 A quick recap of training deep neural networks
71. Deep Learning(CS7015): Lec 9.2 Unsupervised pre-training
72. Deep Learning(CS7015): Lec 9.3 Better activation functions
73. Deep Learning(CS7015): Lec 9.4 Better initialization strategies
74. Deep Learning(CS7015): Lec 9.5 Batch Normalization
75. Deep Learning(CS7015): Lec 10.1 One-hot representations of words
76. Deep Learning(CS7015): Lec 10.2 Distributed Representations of words
77. Deep Learning(CS7015): Lec 10.3 SVD for learning word representations
78. Deep Learning(CS7015): Lec 10.3 (Part-2) SVD for learning word representations (Contd.)
79. Deep Learning(CS7015): Lec 10.4 Continuous bag of words model
80. Deep Learning(CS7015): Lec 10.5 Skip-gram model
81. Deep Learning(CS7015): Lec 10.5 (Part-2) Skip-gram model (Contd.)
82. Deep Learning(CS7015): Lec 10.6 Contrastive estimation
83. Deep Learning(CS7015): Lec 10.7 Hierarchical softmax
84. Deep Learning(CS7015): Lec 10.8 GloVe representations
85. Deep Learning(CS7015): Lec 10.9 Evaluating word representations
86. Deep Learning(CS7015): Lec 10.10 Relation between SVD and Word2Vec
87. Deep Learning(CS7015): Lec 11.1 The convolution operation
88. Deep Learning(CS7015): Lec 11.2 Relation between input size, output size and filter size
89. Deep Learning(CS7015): Lec 11.3 Convolutional Neural Networks
90. Deep Learning(CS7015): Lec 11.3 (Part-2) Convolutional Neural Networks (Contd.)
91. Deep Learning(CS7015): Lec 11.4 CNNs (success stories on ImageNet)
92. Deep Learning(CS7015): Lec 11.4 (Par-2) CNNs (success stories on ImageNet) (Contd.)
93. Deep Learning(CS7015): Lec 11.5 Image Classification continued (GoogLeNet and ResNet)
94. Deep Learning(CS7015): Lec 12.1 Visualizing patches which maximally activate a neuron
95. Deep Learning(CS7015): Lec 12.2 Visualizing filters of a CNN
96. Deep Learning(CS7015): Lec 12.3 Occlusion experiments
97. Deep Learning(CS7015): Lec 12.4 Finding influence of input pixels using backpropagation
98. Deep Learning(CS7015): Lec 12.5 Guided Backpropagation
99. Deep Learning(CS7015): Lec 12.6 Optimization over images
100. Deep Learning(CS7015): Lec 12.7 Create images from embeddings

Only the first 100 are listed; see the full playlist.

Books, notes and websites

- ✅ NPTEL *Deep Learning* (Khapra, IIT Madras) — nptel.ac.in/courses/106106201/
- d2l.ai; Karpathy *Zero to Hero*; Stanford CS231n

- Book: Prince, *Understanding Deep Learning* (free); Goodfellow *Deep Learning*

Practice / proof of understanding

Backprop in NumPy, a ResNet on CIFAR-10, a small transformer language model.

What your IITM course sheet says

From the PDF: DA3409

Official description: : This course aims to introduce undergraduate students to the fundamental concepts of Deep Learning and their practical applications. The course provides a conceptual and hands-on understanding of neural networks and modern deep learning architectures. Students will learn how deep learning models are designed, trained, and applied to solve problems in areas such as computer vision and natural language processing.

Official syllabus (trimmed):

> Foundations of Deep Learning: Brief history and motivation for deep learning. McCulloch–Pitts neuron model, threshold logic, perceptron model, perceptron learning algorithm and convergence. Multilayer perceptrons (MLPs) and the representation power of neural networks. Training Neural Networks: Sigmoid and other activation functions, feedforward neural networks, loss functions, and gradient descent. Automatic differentiation for gradient computation. Gradient-based optimization methods including stochastic gradient descent, momentum, Nesterov accelerated gradient, AdaGrad, RMSProp, and Adam. Regularization and Training Deep Networks: Regularization methods including L2 regularization, early stopping, dataset augmentation, parameter sharing, dropout, and ensemble methods. Techniques for improving training such as better activation functions, weight initialization, and batch/layer normalization. Convolutional Neural Networks: Motivation for convolutional networks. Key architectures such as LeNet, AlexNet, VGG, GoogLeNet, and ResNet. Sequential Models: Recurrent neural networks (RNNs), backpropagation through time, vanishing and exploding gradients. Advanced recurrent architectures including GRU and LSTM. Attention and Modern Architectures: Encoder–decoder models and sequence-to-sequence learning. Attention mechanisms in neural networks. Transformers and self-attentio

Official textbooks: 1. Charu C. Aggarwal. Neural Networks and Deep Learning: A Textbook. Springer. 2019. 2. Aston Zhang, Zachary C. Lipton, Mu Li, Alexander Smola. Dive into Deep Learning

Official references: 1. Ian Goodfellow and Yoshua Bengio and Aaron Courville. Deep Learning. An MIT Press book. 2016. 2. Simon J.D. Prince. Understanding Deep Learning. An MIT Press book. 2023.

From the PDF: DA3410

Official description: The laboratory course is designed to complement the theory course on Deep Learning by providing hands-on experience in implementing and experimenting with neural network models. The course aims to help students understand how deep learning models are trained, evaluated, and improved in practice. Students will learn to implement core deep learning algorithms using modern frameworks such as PyTorch or TensorFlow, work with real world datasets, and gain practical exposure to model training, debugging, and experimentation. The laboratory will emphasize experimentation, visualization, and analysis of model behavior

Official syllabus (trimmed):

> Introduction to Deep Learning Frameworks: Setting up Python environments for deep learning. Introduction to libraries such as PyTorch or JAX. Working with tensors, automatic differentiation, and GPU acceleration. Data Handling and Visualization: Loading and preprocessing datasets. Dataset splitting, normalization, and augmentation. Visualization of data and training statistics using libraries such as matplotlib. Implementation of Basic Neural Networks: Implementing perceptrons and multilayer perceptrons. Building feedforward neural networks with and without using deep learning frameworks. Training models using gradient descent and automatic differentiation.. Optimization and Training Experiments: Experimenting with optimization algorithms such as stochastic gradient descent, momentum, RMSProp, and Adam. Studying learning rate schedules and convergence behavior. Regularization and Model Improvement: Applying techniques such as L2 regularization, dropout, early stopping, and data augmentation. Studying bias variance tradeoff through controlled experiments. Convolutional Neural Networks: Implementing convolutional neural networks for image classification tasks. Training and evaluating CNN models on standard image datasets. Visualizing learned filters and feature maps. Sequential Models: Implementing recurrent neural networks for sequence prediction tasks. Training mod

Official textbooks: 1. Eli Stevens, Luca Antiga, and Thomas Viehmann. Deep Learning with PyTorch, Manning Publications, 2026.

Official references: 1. Ian Goodfellow and Yoshua Bengio and Aaron Courville. Deep Learning. An MIT Press book. 2016. 2. Simon J.D. Prince. Understanding Deep Learning. An MIT Press book. 2023.

From the PDF: DA6401

Official description: 1. Understanding the Fundamentals of Deep Learning: o To provide a comprehensive introduction to the core concepts and mathematical foundations of deep learning. o To equip students with an understanding of key neural network architectures, optimization techniques, and the learning process. 2. Developing Practical Knowledge: o To introduce students to advanced deep learning techniques, including Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Long Short-Term Memory Networks (LSTMs), and Transformer models. 3. Implementation and Analysis: o To enable students to implement deep learning algorithms using frameworks like PyTorch or JAX. o To evaluate the performance of neural networks and optimize them for various applications.

Official syllabus (trimmed):

1. Introduction and History of Deep Learning (3 Lectures): o Overview of deep learning and its evolution. o Success stories in modern AI.
2. Neural Network Fundamentals (6 Lectures): o McCulloch Pitts Neuron, Thresholding Logic, Perceptrons. Perceptron Learning Algorithm, Multilayer Perceptrons (MLPs), Representation Power of MLPs
3. Feedforward Neural Networks (6 Lectures): o Sigmoid Neurons, Feedforward Neural Networks.Representation Power of Feedforward Networks, automatic differentiation, stochastic optimisation.
4. Optimization Techniques (6 Lectures): o Gradient Descent (GD), Momentum-based GD, Nesterov Accelerated GD, Stochastic GD, AdaGrad, RMSProp, Adam.
5. Regularization Techniques (4 Lectures): o Bias-Variance Tradeoff, L2 Regularization, Early Stopping, Dataset Augmentation. Ensemble Methods, Dropout, and Injecting Noise.
6. Advanced Neural Network Training Methods (4 Lectures): o Greedy Layerwise Pre-training, Improved Activation Functions, Batch Normalization.
7. Convolutional Neural Networks (CNNs) (3 Lectures): o LeNet, AlexNet, ZFNet, VGGNet, GoogLeNet, ResNet.
8. Word Embeddings and Vector Representations (3 Lectures): o Introduction to Word2Vec and Skip-gram models.
9. Recurrent Neural Networks (RNNs) (4 Lectures): o Computations in a RNN, Backpropagation Through Time (BPTT), Vanishing and Exploding Gradients. o Long Short-Term Memory C

Official textbooks: 1. Charu C. Aggarwal. Neural Networks and Deep Learning: A Textbook. Springer. 2019. 2. Aston Zhang, Zachary C. Lipton, Mu Li, Alexander Smola. Dive into Deep Learning.

Official references: 1. Ian Goodfellow and Yoshua Bengio and Aaron Courville. Deep Learning. An MIT Press book. 2016. 2. Simon J.D. Prince. Understanding Deep Learning. An MIT Press book. 2023.